Bayesian Linear Model Preliminary

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Outline

Some basic notions and algorithms

Bayesian Normal Linear Model

Bayesian Linear Mixed Model Basis

Comparision of multiple LMM models

Future work

References

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Bayesian theorem

$$p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\mathbf{y}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})}{p(\mathbf{y})} \tag{1}$$

Usefully, one can get

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto p(\mathbf{y}|\boldsymbol{\theta}) \times \pi(\boldsymbol{\theta}) \tag{2}$$

or

$$\textit{Posterior} \propto \textit{Likelihood} \times \textit{Prior} \tag{3}$$

Scale-inv-χ^2 distribution and Inv-Gamma distribution

Scale-inv-χ^2 distribution:

$$f(x;\nu,\tau^2) = \frac{(\tau^2\nu/2)^{\nu/2}}{\Gamma(\nu/2)}\frac{exp(\frac{-\nu\tau^2}{2x})}{x^{1+\nu/2}} \tag{4}$$

Inv-Gamma distribution:

$$f(x,\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\frac{exp(\frac{-\beta}{x})}{x^{\alpha+1}} \tag{5}$$

So

$$x \sim_{\textbf{Scaled Inv}} -_{\chi}{}^2(\nu,\tau^2) \equiv x \sim_{\textbf{Inv-Gamma}}(\alpha = \frac{\nu}{2}, \beta = \frac{\nu\tau^2}{2}) \tag{6}$$

Metropolis-Hastings algorithm

Based on Accept-reject theory and Markov Chain theory[(1,2)]

Suitable to get samples from complex distributions.

Algorithm 1: Metropolis-Hastings Sampling

Initialization: Set the arbitrary value $x^{(0)}$

for $j \leftarrow 1$ *to* N **do**

  Genearte y from $q(x^{(j)}, .)$ and u from $U(0,1)$

  if $u \leq \alpha(x^{(j)}, y) = min[1, \frac{\pi(y)q(y,x^{(j)})}{\pi(x^{(j)})q(x^{(j)},y)}]$ **then**

    Set $x^{(j)} = y$

  else

    Set $x^{(j+1)} = x^{(j)}$;

  end

end

Result: Return the values $\{x^{(1)}, x^{(2)}, ..., x^{(N)}\}$

Gibbs algorithm

Suitable to high-dimensional distribution sampling[(3)]

Algorithm 2: Gibbs Sampling

Initialization: Set the arbitrary **vector** value
$\mathbf{x}^{(0)} = (x_1^{(0)}, x_2^{(0)}, x_3^{(0)}, ..., x_n^{(0)})$
for $i \leftarrow 1$ *to* N **do**
  Draw Sequence $\mathbf{x}^{(i)}$:
  // Sampling based on full conditional distribution
  $x_1^{(i)}$ from $f(x_1^{(i)}|x_2^{(i)}, x_3^{(i)}, ..., x_n^{(i)})$
  $x_2^{(i)}$ from $f(x_2^{(i)}|x_1^{(i)}, x_3^{(i)}, ..., x_n^{(i)})$
  ...
  $x_j^{(i)}$ from $f(x_j^{(i)}|x_1^{(i)}, x_2^{(i)}, ..., x_{j-1}^{(i)}, x_{j+1}^{(i)}, ..., x_n^{(i)})$
  ...
  $x_n^{(i)}$ from $f(x_n^{(i)}|x_1^{(i)}, x_2^{(i)}, ..., x_j^{(i)}, ..., x_{n-1}^{(i)})$
end
Result: Return a Markov chain $\{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, ..., \mathbf{x}^{(N)}\}$ with stationary $f(x_1, x_2, ..., x_n)$.

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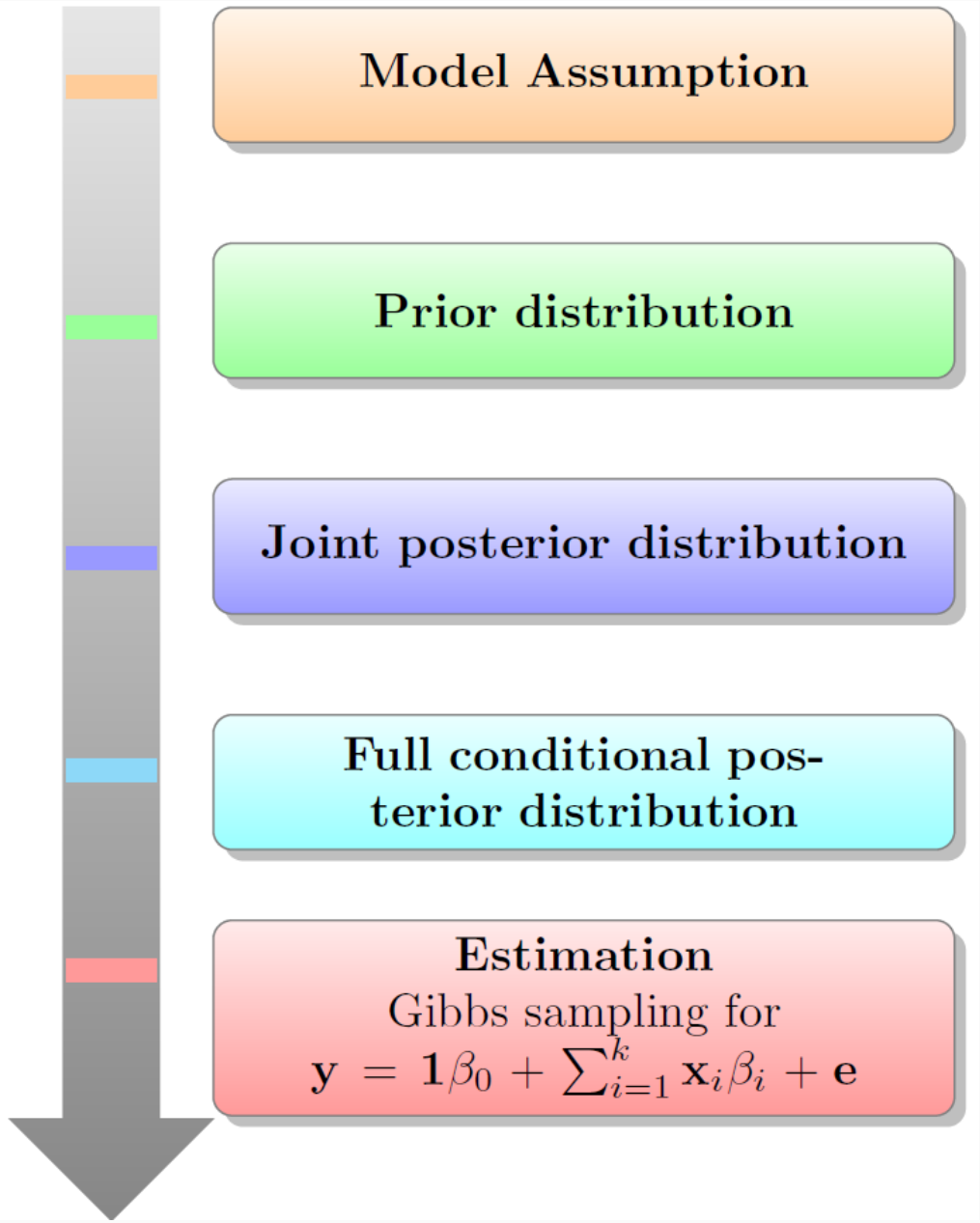
Common Procedure



Model

Normal linear model:

$$y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \mathbf{e}_i, i = 1, ..., n. \tag{7}$$

where $\boldsymbol{\beta}$ is a $k \times 1$ vector, $\mathrm{e_i} \sim \mathrm{N}(0,\sigma^2)$.

Conditional distribution:

$$p(\mathbf{y}|\boldsymbol{\beta}, \sigma^2) \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \mathbf{I}\sigma^2) \tag{8}$$

Corresponding likelihood function:

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\beta}, \sigma^2) \propto (\sigma^2)^{-n/2} exp(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})) \tag{9}$$

Where $\mathbf{y} = (y_1, y_2, ..., y_n)^T$, $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_n)^T$.

Prior distribution

Rewrite Likelihood function(9):

$$
\begin{aligned}
p(\mathbf{y}|\mathbf{X}, \boldsymbol{\beta}, \sigma^2) &\propto (\sigma^2)^{-n/2} exp((\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T(\mathbf{X}^T\mathbf{X})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) \\
&= (\sigma^2)^{-n/2} exp(-\frac{\nu s^2}{2\sigma^2})(\sigma^2)^{-(n-v)/2} exp(-\frac{1}{2\sigma^2}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T(\mathbf{X}^T\mathbf{X})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}))
\end{aligned} \tag{10}
$$

Where $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$,$\nu s^2 = (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$,
$\nu = n - k$, and k is the number of regression coefficients.
Equation(10) suggests:

$$
\begin{aligned}
p(\boldsymbol{\beta}, \sigma^2) &= p(\sigma^2)p(\boldsymbol{\beta}|\sigma^2) \\
p(\sigma^2) &\sim \textbf{Scale-Inv-}\chi^2(\nu_0, s_0^2) \\
or \quad p(\sigma^2) &\sim \textbf{Inv-Gamma}(a_0 = \nu_0/2, b_0 = (\nu_0 s_0^2)/2) \\
p(\boldsymbol{\beta}|\sigma^2) &\sim \mathcal{N}(\mu_0, \sigma^2\boldsymbol{\Lambda}_0^{-1})
\end{aligned} \tag{11}
$$

Joint posterior distribution

Joint posterior distribution[(4)] is,

$$
\begin{aligned}
p(\boldsymbol{\beta}, \sigma^2|\mathbf{y}) &\propto p(\mathbf{y}|\boldsymbol{\beta}, \sigma^2)p(\boldsymbol{\beta}|\sigma^2)p(\sigma^2) \\
&\propto \frac{1}{(\sigma^2)^{n/2}} exp(-\frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})^T(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}))\frac{1}{(\sigma^2)^{k/2}} exp(-\frac{1}{2\sigma^2}(\boldsymbol{\beta}-\mu_0)^T\boldsymbol{\Lambda}_0(\boldsymbol{\beta}-\mu_0))\frac{1}{(\sigma^2)^{a_0+1}} exp(-\frac{b_0}{\sigma^2}) \\
&= \frac{1}{(\sigma^2)^{n/2}}\frac{1}{(\sigma^2)^{k/2}}\frac{1}{(\sigma^2)^{a_0+1}} exp(-\frac{b_0}{\sigma^2}) exp(-\frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})^T(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}) - \frac{1}{2\sigma^2}(\boldsymbol{\beta}-\boldsymbol{\mu}_0)^T\boldsymbol{\Lambda}_0(\boldsymbol{\beta}-\boldsymbol{\mu}_0)) \\
&= \frac{1}{(\sigma^2)^{n/2}}\frac{1}{(\sigma^2)^{k/2}}\frac{1}{(\sigma^2)^{a_0+1}} exp(-\frac{b_0}{\sigma^2}) exp(-\frac{1}{2\sigma^2}(\mathbf{y}^T\mathbf{y} + \boldsymbol{\beta}^T\underbrace{(\mathbf{X}^T\mathbf{X}+\boldsymbol{\Lambda}_0)}_{\boldsymbol{\Lambda_n}}\boldsymbol{\beta} - 2\boldsymbol{\beta}^T(\mathbf{X}^T\mathbf{y}+\boldsymbol{\Lambda}_0\boldsymbol{\mu}_0) + \boldsymbol{\mu}_0^T\boldsymbol{\Lambda}_0\boldsymbol{\mu}_0)) \\
&= \frac{1}{(\sigma^2)^{n/2}}\frac{1}{(\sigma^2)^{k/2}}\frac{1}{(\sigma^2)^{a_0+1}} exp(-\frac{b_0}{\sigma^2}) exp(-\frac{1}{2\sigma^2}(\mathbf{y}^T\mathbf{y} + \boldsymbol{\beta}^T\boldsymbol{\Lambda_n}\boldsymbol{\beta} - 2\boldsymbol{\beta}^T\boldsymbol{\Lambda_n}\underbrace{\boldsymbol{\Lambda_n^{-1}}(\mathbf{X}^T\mathbf{y}+\boldsymbol{\Lambda}_0\boldsymbol{\mu}_0)}_{\boldsymbol{\mu_n}} + \boldsymbol{\mu}_0^T\boldsymbol{\Lambda}_0\boldsymbol{\mu}_0)) \\
&= \frac{1}{(\sigma^2)^{n/2}}\frac{1}{(\sigma^2)^{k/2}}\frac{1}{(\sigma^2)^{a_0+1}} exp(-\frac{b_0}{\sigma^2}) exp(-\frac{1}{2\sigma^2}(\mathbf{y}^T\mathbf{y} + \boldsymbol{\beta}^T\boldsymbol{\Lambda_n}\boldsymbol{\beta} - 2\boldsymbol{\beta}^T\boldsymbol{\Lambda_n}\boldsymbol{\mu_n} + \boldsymbol{\mu}_0^T\boldsymbol{\Lambda}_0\boldsymbol{\mu}_0)) \\
&= \frac{1}{(\sigma^2)^{n/2}}\frac{1}{(\sigma^2)^{k/2}}\frac{1}{(\sigma^2)^{a_0+1}} exp(-\frac{b_0}{\sigma^2}) exp(-\frac{1}{2\sigma^2}(\mathbf{y}^T\mathbf{y} + (\boldsymbol{\beta}-\boldsymbol{\mu_n})^T\boldsymbol{\Lambda_n}(\boldsymbol{\beta}-\boldsymbol{\mu_n}) - \boldsymbol{\mu}_n^T\boldsymbol{\Lambda}_n\boldsymbol{\mu}_n + \boldsymbol{\mu}_0^T\boldsymbol{\Lambda}_0\boldsymbol{\mu}_0)) \\
&= \frac{1}{(\sigma^2)^{k/2}} exp(-\frac{1}{2\sigma^2}(\boldsymbol{\beta}-\boldsymbol{\mu_n})^T\boldsymbol{\Lambda_n}(\boldsymbol{\beta}-\boldsymbol{\mu_n}))\underbrace{\frac{1}{(\sigma^2)^{n/2}}\frac{1}{(\sigma^2)^{a_0+1}}}_{(1/(\sigma^2))^{a_n+1}} exp(-\frac{1}{\sigma^2}(\underbrace{\frac{1}{2}(\mathbf{y}^T\mathbf{y} - \boldsymbol{\mu}_n^T\boldsymbol{\Lambda}_n\boldsymbol{\mu}_n + \boldsymbol{\mu}_0^T\boldsymbol{\Lambda}_0\boldsymbol{\mu}_0) + b_0}_{b_n})) \\
&\propto p(\boldsymbol{\beta}|\boldsymbol{\mu}_n, \boldsymbol{\Lambda}_n^{-1}\sigma^2)p(\sigma^2|\mathbf{Y})
\end{aligned}
\tag{12}
$$

Full conditional posterior distribution

$$
\begin{aligned}
p(\boldsymbol{\beta}|\mathbf{y}, \mathbf{X}, \sigma^2) &\sim \mathcal{N}(\boldsymbol{\mu}_n, \boldsymbol{\Lambda}_n^{-1}\sigma^2) \\
p(\sigma^2|\boldsymbol{\beta}, \mathbf{y}, \mathbf{X}) &\sim \textbf{Inv-Gamma}(a_n, b_n)
\end{aligned}
\tag{13}
$$

Where,

$$
\begin{aligned}
\boldsymbol{\mu}_n &= \boldsymbol{\Lambda}_{\boldsymbol{n}}^{-\mathbf{1}}(\mathbf{X}^T\mathbf{y} + \boldsymbol{\Lambda}_0\boldsymbol{\mu}_0) \\
\boldsymbol{\Lambda}_n &= \mathbf{X}^T\mathbf{X} + \boldsymbol{\Lambda}_0, \\
a_n &= a_0 + n/2 \\
b_n &= b_0 + \frac{1}{2}(\mathbf{y}^T\mathbf{y} - \boldsymbol{\mu}_n^T\boldsymbol{\Lambda}_n\boldsymbol{\mu}_n + \boldsymbol{\mu}_0^T\boldsymbol{\Lambda}_0\boldsymbol{\mu}_0)
\end{aligned}
\tag{14}
$$

Full conditional posterior distribution

Assuming improper prior distribution [1] for $\boldsymbol{\beta}$:
$(p(\boldsymbol{\beta}) \propto constant)$

$$p(\boldsymbol{\beta}, \sigma^2|\mathbf{y}) \propto \underbrace{\frac{1}{(\sigma^2)^{n/2}} \frac{1}{(\sigma^2)^{a_0+1}}}_{(1/(\sigma^2))^{a_n+1}} exp(-\frac{1}{\sigma^2} \underbrace{(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})}_{b_n})$$
$$\sim \textbf{Inv-Gamma}(a_n, b_n) \tag{15}$$

Assuming ridge regression: $\mathbf{u}_0 = 0, \boldsymbol{\Lambda}_0 = \lambda\mathbf{I}$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})\mathbf{X}^T\mathbf{y} \tag{16}$$

[1] meaning that it does not integrate to 1.

Gibbs sampling for $\mathbf{y} = \mathbf{1}\beta_0 + \sum_{i=1}^{k} \mathbf{x}_i\beta_i + \mathbf{e}$

```
##########################################################################################
set.seed(1234)
n = 2000; p = 10            # number of observations; covariates
ve = 5; vb = 3               # residual variance;  prior variance for slopes
x = matrix(sample(c(0,1,2),n*k,replace=T),nrow=n,ncol=k)
X = cbind(1,x)
(betaTrue = c(1,rnorm(k,0,sqrt(vb)))) # β_True = (β_1, ..., β_{k+1}), where β_2, .., β_{k+1} from N(0, Iσ_b^2).
y = X%*%betaTrue + rnorm(n,0,sqrt(ve))  # y = Xβ + e, where e from N(0, Iσ_e^2).
lambda = ve/vb      # λ = σ_e^2/σ_b^2
mme = t(X)%*%X     # X^T X
diag(mme[-1,-1]) = diag(mme[-1,-1]) + lambda # (X^T X)_ii = (X^T X)_ii + λ, where i = 2, .., 11.
rhs = t(X)%*%y; mmeInv = solve(mme); (mmeSoln = mmeInv%*%rhs)
beta=c(mean(y),array(0,k))      # starting β = (β_1, β_2, ..., β_{k+1})^T = (ȳ, 0, ..., 0)^T
ve = 1   # starting σ̂_e^2 = 1
# loop for Gibbs sampler
niter = 10000 # number of samples
samples = matrix(0,niter,k+1)
meanBeta = array(0,k+1)    # β̄ = (0, .., 0)^T
meanVe = 0      # σ̄_e^2 = 0
for (iter in 1:niter){
|   # 1.Sampling intercept
|   w = y - X[, -1] %*% beta[-1]    # w = y − (x_2, ..., x_{k+1})(β_2, ..., β_{k+1})^T
|   x = X[, 1]
|   xpxi = 1/(t(x)%*%x)     # (x_1^T x_1)^{-1}
|   betaHat = t(x)%*%w*xpxi    # β̂ = x_1^T w · (x_1^T x_1)^{-1}
|   beta[1] = rnorm(1,betaHat,sqrt(xpxi*ve))    # Get β_1 from N(β̂, (x_1^T x_1)^{-1} · σ̂_e^2)
|   lambda = ve/vb     # λ = V_e/V_b
|   # 2.Sampling slopes
|   for (j in 2:(k+1)){
|   |   # w = y − (x_1, ..., x_{j−1}, x_{j+1}, ..., x_{k+1})(β_1, ..., β_{j−1}, β_{j+1}, ..., β_{k+1})^T
|   |   w = y - X[,-j]%*%beta[-j]
|   |   x = X[,j]
|   |   xpxi = 1/(t(x)%*%x + lambda)  # (x_j^T x_j)^{-1} + λ
|   |   betaHat = t(x)%*%w*xpxi                    # β̂ = x_j^T w · ((x_j^T x_j)^{-1} + λ)
|   |   beta[j] = rnorm(1,betaHat,sqrt(xpxi*ve))  # Get β_2 from N(β̂, (x_j^T x_j)^{-1} · σ̂_e^2)
|   }
|   meanBeta = meanBeta + beta  # β̄ = β̄ + β
|   samples[iter,] = beta
|   # 3.Sampling residual var
|   nu = 4.2
|   scale = 2.5
|   # σ_e^2|y, β ∼ Inv-Gamma(a_n, b_n) ≡ Scale-Inv-χ^2(ν_0 + n/2, ν_0 s_0^2 + (y/2 − Xβ)^T (y − Xβ)/2)
|   ve = (t(y - X %*% beta) %*% (y - X %*% beta) + nu*scale)/rchisq(1,n+nu)
|   meanVe = meanVe + ve    # σ̄_e^2 = σ̄_e^2 + σ̂_e^2
|   if ((iter%%1000)==0){
|   |   cat (sprintf("\n Intercept = %6.3f \n", meanBeta[1]/iter))
|   |   for (j in 2:(k+1)){
|   |   |   cat (sprintf("Slope[%2d] = %6.3f \n", j-1, meanBeta[j]/iter))
|   |   }
|   }
}
meanVe = meanVe[1,1]/niter
##########################################################################################
```

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Linear mixed model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \sum_{i=1}^{m} \mathbf{Z}_i \mathbf{u}_i + \mathbf{e} \tag{17}$$

Where $\mathbf{y}$ is $n \times 1$ data vector, $\boldsymbol{\beta}$ is $p \times 1$ fixed effect vector with corresponding n matrix $\mathbf{X}$, $\mathbf{u}_i$ is $q_i \times 1$ random vector with corresponding $n \times q_i$ matrix $\mathbf{Z}_i$, $\mathbf{e}$ is $n \times 1$ random residuals vector.

Conditional distribution:

$$\begin{aligned} p(\mathbf{u}_i | \mathbf{G}_i, \sigma^2_{\mathbf{u}_i}) &\sim \mathcal{N}_{q_i}(\mathbf{0}, \mathbf{G}_i \sigma^2_{\mathbf{u}_i}), i = 1, ..., m, \\ p(\boldsymbol{e} | \sigma^2_e) &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}\sigma^2_e), \\ p(\mathbf{y} | \boldsymbol{\beta}, \mathbf{u}_1, ..., , \mathbf{u}_m, \sigma^2_e) &\sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta} + \sum_{i=1}^{m} \mathbf{Z}_i \mathbf{u}_i, \mathbf{I}\sigma^2_e). \end{aligned} \tag{18}$$

Prior distribution[(5)]

Improper prior for $\boldsymbol{\beta}$:

$$p(\boldsymbol{\beta}) \propto constant. \tag{19}$$

Prior for σ_e^2:

$$p(\sigma_e^2|\nu_e, s_e^2) \propto (\sigma_e^2)^{-(\nu_e/2+1)} exp(-\frac{\nu_e s_e^2}{2\sigma_e^2}) \tag{20}$$

Prior for $\sigma_{\mathbf{u}_i}^2$:

$$p(\sigma_{\mathbf{u}_i}^2|\nu_{\mathbf{u}_i}, s_{\mathbf{u}_i}^2) \propto (\sigma_{\mathbf{u}_i}^2)^{-(\nu_{\mathbf{u}_i}/2+1)} exp(-\frac{\nu_{\mathbf{u}_i} s_{\mathbf{u}_i}^2}{2\sigma_{\mathbf{u}_i}^2}), i = 1, ..., m \tag{21}$$

Joint posterior distribution

$$
\begin{aligned}
p(\boldsymbol{\beta}, \mathbf{u}_1, \ldots, \mathbf{u}_m, \sigma^2_{\mathbf{u}_1}, \ldots, \sigma^2_{\mathbf{u}_m}, \sigma^2_e | \mathbf{y}) \propto\ & p(\mathbf{y}|\boldsymbol{\beta}, \mathbf{u}_1, \ldots, \mathbf{u}_m, \sigma^2_{\mathbf{u}_1}, \ldots, \sigma^2_{\mathbf{u}_m}, \sigma^2_e) \\
& \cdot p(\mathbf{u}_1, \ldots, \mathbf{u}_m | \boldsymbol{\beta}, \sigma^2_{\mathbf{u}_1}, \ldots, \sigma^2_{\mathbf{u}_m}, \sigma^2_e) \\
& \cdot p(\sigma^2_{\mathbf{u}_1}, \ldots, \sigma^2_{\mathbf{u}_m} | \boldsymbol{\beta}, \sigma^2_e) \\
& \cdot p(\sigma^2_e | \boldsymbol{\beta}) \\
& \cdot p(\boldsymbol{\beta}) \\
\propto\ & (\sigma^2_e)^{-n/2} exp(-\frac{1}{2\sigma^2_e}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \sum_{i-1}^{m} \mathbf{Z}_i \mathbf{u}_i)^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \sum_{i-1}^{m} \mathbf{Z}_i \mathbf{u}_i)) \\
& \cdot \prod_{i=1}^{m} ((\sigma^2_{\mathbf{u}_i})^{-q_i/2} exp(-\frac{1}{2\sigma^2_{\mathbf{u}_i}} \mathbf{u}_i^T \mathbf{G}_i^{-1} \mathbf{u}_i)) \\
& \cdot \prod_{i=1}^{m} ((\sigma^2_{\mathbf{u}_i})^{-(\nu_{\mathbf{u}_i}/2+1)} exp(-\frac{1}{2\sigma^2_{\mathbf{u}_i}} \nu_{\mathbf{u}_i} s^2_{\mathbf{u}_i})) \\
& \cdot (\sigma^2_e)^{-(\nu_e/2+1)} exp(-\frac{1}{2\sigma^2_e} \nu_e s^2_e) \\
& \cdot constant
\end{aligned}
\tag{22}
$$

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Frequentist LMM: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \mathbf{e}$

- Assumption[(6,7)]: All loci have equal variance σ^2.

L_0 $\quad Y = X\beta + Zu + e$

$\eta = \mu \mid u$

L_1 $\quad y \sim N(\mu, V) \qquad \hat{\beta}_{REML} = (X^T V^{-1} X)^{-1} X^T V^{-1} Y \qquad u_i \sim N(0, \sigma^2) \qquad e \sim N(0, I\sigma_e^2)$

L_2 $\quad \mu = X\beta \qquad \hat{u}_{BLUP} = GZV^{-1}(Y - X\beta)$

L_3 $\quad V = ZGZ^T + R$

BayesA: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \sum_{j=1}^{m} \mathbf{Z}_j u_j + \mathbf{e}$

- Assumption[(7)]: Prior of variance of chr seg $Var(g_{ij}) = \sigma^2_{gi}$.

L_0 $$Y = X\beta + \sum_{i=1}^{m} Z_j u_j + e$$

L_1 $$\beta_{\text{Prior}} \qquad u_j \sim N(0, \sigma_j^2) \qquad e_i \sim N(0, I\sigma_e^2)$$

L_2 $$\sigma_j^2 \sim \nu_\alpha S_\alpha^2 \chi_{\nu_\alpha}^{-2} \qquad \sigma_e^2 \sim \nu_e S_e^2 \chi_{\nu_e}^{-2}$$

L_3 $$S_\alpha^2 = \frac{V_A}{2\overline{pq}(1-\pi)m} \frac{\nu_\alpha - 2}{\nu_\alpha} \qquad S_e^2 = \frac{V_E(\nu_e - 2)}{\nu_e}$$

L_4 $$h_a^2 = \frac{V_A}{V_P} = \frac{V_P - V_E}{V_P}$$

BayesB: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \sum_{j=1}^{m} \mathbf{Z}_j \alpha_j \delta_j + \mathbf{e}$

- Assumption[(7,8)]: Adding density peak π at $\sigma^2_{gi} = 0$

L_0 $\quad Y = X\beta + \sum_{i=1}^{m} Z_j u_j + e$

L_1 $\quad \beta_{\text{Prior}} \qquad \alpha_j \bullet \delta_j \qquad e_i \sim N(0, I\sigma_e^2)$

L_2 $\quad \alpha_j \mid \sigma_j^2 \sim N(0, \sigma_j^2) \qquad \delta_j = \begin{cases} 1 & p = 1-\pi \\ 0 & p = \pi \end{cases} \qquad \sigma_e^2 \sim \nu_e S_e^2 \chi_{\nu_e}^{-2}$

L_3 $\quad \alpha_j \sim t_{\nu_\alpha}(0, S_\alpha^2) \qquad \sigma_j^2 \sim \nu_\alpha S_\alpha^2 \chi_{\nu_\alpha}^{-2} \qquad \pi = \frac{m - s \cdot k}{m}$

L_4 $\quad S_\alpha^2 = \frac{V_A}{2\overline{pq}(1-\pi)m} \frac{\nu_\alpha - 2}{\nu_\alpha} \qquad S_e^2 = \frac{V_E(\nu_e - 2)}{\nu_e}$

L_5 $\quad h_a^2 = \frac{V_A}{V_P} = \frac{V_P - V_E}{V_P}$

BayesC: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \sum_{j=1}^{m} \mathbf{Z}_j \alpha_j \delta_j + \mathbf{e}$

- Assumption[(9)]: $\alpha_j | \sigma_\alpha^2 \sim N(\mu,\ \sigma_\alpha^2)$

L_0 $\quad Y = X\beta + \sum_{i=1}^{m} Z_j u_j + e$

L_1 $\quad \beta_{\text{Prior}} \qquad \alpha_j \bullet \delta_j \qquad e \sim N(0, I\sigma_e^2)$

L_2 $\quad \alpha_j | \sigma_\alpha^2 \sim N(0, \sigma_\alpha^2) \qquad \delta_j = \begin{cases} 1 & p = 1-\pi \\ 0 & p = \pi \end{cases} \qquad \sigma_e^2 \sim \nu_e S_e^2 \chi_{\nu_e}^{-2}$

L_3 $\quad \alpha_j \sim t_{\nu_\alpha}(0, S_\alpha^2) \qquad \pi = \frac{m - s \cdot k}{m}$

L_4 $\quad S_\alpha^2 = \frac{V_A}{2\overline{pq}(1-\pi)m} \frac{\nu_\alpha - 2}{\nu_\alpha} \qquad S_e^2 = \frac{V_E(\nu_e - 2)}{\nu_e}$

L_5 $\quad h_a^2 = \frac{V_A}{V_P} = \frac{V_P - V_E}{V_P}$

BayesN: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \sum_{i=1}^{w}\sum_{j=1}^{m_i} \mathbf{Z}_{ij}\alpha_{ij}\delta_{ij}\Delta_i + \mathbf{e}$

- Assumption[(8)] : Δ_j indicate the window in the model.

L_0

$$Y = X\beta + \sum_{i=1}^{m} Z_j u_j + e$$

L_1

$$\beta_{\text{Prior}} \qquad \alpha_{ij} \bullet \delta_{ij} \bullet \Delta_i \qquad e \sim N(0, I\sigma_e^2)$$

L_2

$$\alpha_{ij} \mid \sigma_i^2 \sim N(0, \sigma_i^2) \qquad \delta_{ij} = \begin{cases} 1 & p = 1-\pi \\ 0 & p = \pi \end{cases} \qquad \Delta_j = \begin{cases} 1 & p = 1-\Pi \\ 0 & p = \Pi \end{cases} \qquad \sigma_e^2 \sim \nu_e S_e^2 \chi_{\nu_e}^{-2}$$

L_3

$$\boldsymbol{\alpha}_i \sim t_{\nu_\alpha}(\mathbf{0}, \mathbf{I}S_\alpha^2) \qquad \sigma_i^2 \sim \nu_\alpha S_\alpha^2 \chi_{\nu_\alpha}^{-2} \qquad \pi = \frac{m - s \cdot k}{m} \qquad \Pi = \frac{w - s}{w}$$

L_4

$$S_\alpha^2 = \frac{V_A}{2\overline{pq}(1-\pi)m} \frac{\nu_\alpha - 2}{\nu_\alpha} \qquad S_e^2 = \frac{V_E(\nu_e - 2)}{\nu_e}$$

L_5

$$h_a^2 = \frac{V_A}{V_P} = \frac{V_P - V_E}{V_P}$$

Outline

Some basic notions and algorithms

Bayesian Normal Linear Model

Bayesian Linear Mixed Model Basis

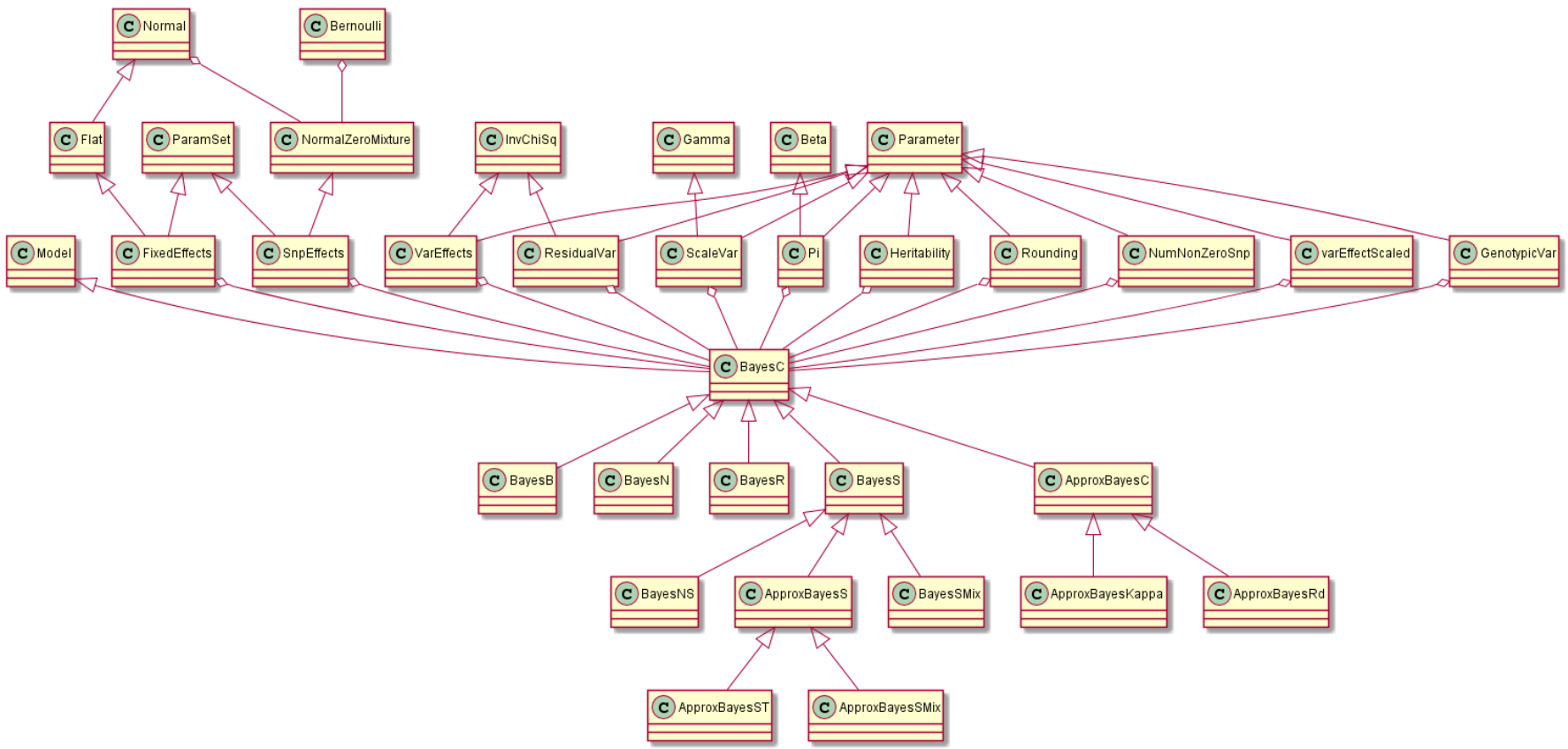
Comparision of multiple LMM models

Future work

References

UML for GCTB Bayesian models

Over 13 Bayesian models[2] .



[2]The solid triangle points to the parent class,Generalization

Solid diamond precision correlation is inseparable,Aggregation.

Outline

Some basic notions and algorithms

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